Yanlong Zhao

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Bio

I am a doctoral student in the **School of Data and Information Sciences at the University of North Carolina at Chapel Hill**. My research lies at the intersection of **machine learning, computational oncology, and precision medicine**.

I currently study **biomarkers of chemotherapy response and resistance in small cell lung cancer (SCLC)** and develop methods for **learning from similar patients and preclinical models**. My broader goal is to build reliable and interpretable learning systems that integrate molecular profiles, drug response, and functional dependency data to support precision oncology.

Education

University of North Carolina at Chapel Hill, Ph.D. Student in Data and Information Science — Present

University of Rochester, M.S. in Electrical and Computer Engineering — Aug. 2023–May 2025

Shandong University, B.E. in Automation — Sep. 2019–Jun. 2023

Research Experience

University of North Carolina at Chapel Hill, Graduate Researcher – Chapel Hill, NC — Present
Advisor: Prof. Jun Li
Investigate multimodal biomarkers of chemotherapy response and resistance in SCLC and develop similarity-based methods that connect patients with biologically relevant patient cohorts and preclinical models for treatment-response estimation.

Prof. Ren Wang@ TIML Lab (Co-advised by Prof. Can Chen), Research Assistant — Jul. 2024–Dec. 2025
Developed graph and hypergraph learning methods that integrate network topology, molecular representations, and biomedical domain knowledge for data-driven discovery. This work produced computational frameworks for drug repositioning and genome-scale metabolic model reconstruction.

Publications

A Multi-Way SMILES-Based Hypergraph Inference Network for Metabolic Model Reconstruction — 2026

Yanlong Zhao*, Yixiao Chen*, Yi Yu, Xiang Liu, Jiawen Du, Jun Wen, Quan Sun, Ren Wang, and Can Chen
Communications Biology, 9, Article 531. DOI

Dual-Route Embedding-Aware Graph Neural Networks for Drug Repositioning — 2025

Yanlong Zhao, Yixiao Chen, Jiawen Du, Jun Wen, Quan Sun, Ren Wang, and Can Chen
Briefings in Bioinformatics, 26(5), bbaf555. DOI

Teaching Experience

DATA 110 @ University of North Carolina at Chapel Hill, Teaching Assistant — 2026 Fall

Industry Experience

DataOceanAI, Project Assistant – San Jose, CA — Nov. 2025–Jun. 2026

Support large-scale machine-learning data operations involving data collection, curation, quality control, and delivery.